
Intelligence Is Not the Bottleneck: Structural Barriers to Automating Alignment Research

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Abstract

Despite the potential for catastrophic harms from human-level AI models, it is not yet clear how to align AI to human interests robustly. Frontier AI developers have proposed to solve this alignment problem using automated alignment research. One version of this claim, which we refer to as the *big* version, assumes that alignment research productivity is capability-bottlenecked and will thus scale alongside the risks posed by more advanced AI capabilities. Based on the big claim, some researchers and forecasters have argued that there will be a critical period of 6-12 months in which most alignment work will occur. In this position paper, we argue that this big claim makes a critical error. We observe that making beneficial progress on automated alignment depends on building institutional capacity and argue that structural barriers, not intelligence, are the principal bottleneck to conducting meaningful automated alignment research.

1. Introduction

Position

Despite the ways in which AI will meaningfully assist alignment research, no realistic amount of alignment research automation will be sufficient to reduce catastrophic risks without substantial institutional progress.

A widely held view in the AI alignment community holds that sufficiently capable AI systems, developed before dangerously capable AGI arrives, could accelerate alignment research enough to address remaining misalignment risks.

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A modest version of this is plausible: AI assistance already shows real benefits for certain categories of alignment-relevant work. For example, AI developers and evaluators regularly use AI models to assist in red-teaming and flaw discovery in models (Mazeika et al., 2024).

However, there is also a bigger, more ambitious version of that claim. We summarize it as: *When AI research and development can be fully automated, datacenters full of research agents will compress a decade's worth of alignment research progress into 6-12 months. This progress can outpace the risks from recursively self-improving AI capabilities in the same period.* (Wiblin, 2026; Dario Amodei, 2024; Leopold Aschenbrenner, 2024). The arguments for this claim assume that the most dangerous AI capabilities will be unlocked by a period of recursive AI self-improvement.

That version of the claim comes with potentially enormous strategic implications. If accurate, it would imply that current, human-driven alignment research efforts will be much less impactful than equivalent effort spent in the automated research era. Likewise, the risks of building AGI look less serious if AI can be trusted to align itself. If the bottleneck in alignment is raw cognitive throughput, the important question would be how that throughput should be allocated.

Although that question is indeed important, we argue that the challenge is even greater. Deploying broadly capable AI at scale would accelerate the most easily verifiable research first, pulling it further ahead of the structural work that is still necessary for AI to be aligned and safe.

2. AI automates data generation but not data review

The big claim rests on an implicit assumption about the compressibility of alignment progress: that the unsolved parts are continuous with the parts AI has already accelerated. This suggests that more automated research capacity translates into faster progress on automated alignment.

However, the distribution of accelerability is not uniform. AI most readily accelerates the technical and formal dimensions of alignment work—the parts with clear metrics, fast feedback loops, and well-defined objectives (Sayash Kapoor

et al., 2026). These traits are absent from many major open problems in alignment, which include questions of who should govern AI and how, which values AI should abide by, and how to empower humans equitably (Ji et al., 2023).

In the analogous context of clinical drug development, AI-originated drug candidates can be generated quickly, with unprecedented success rates in Phase I clinical trials. Despite this fact, AI-originated drugs show only average success in Phase II trials. Since the regulatory approval process is bottlenecked by timelines for recruiting and monitoring patients, the surge in AI-originated drugs has yet to translate to real patient outcomes (Emilia Javorsky, 2026). The success of AI alignment research has a comparable dependence on its adoption and long-term human outcomes.

3. Human-driven institutions are intrinsically important

Someone has to decide whom AI is aligned to. “Alignment” does not exist in a vacuum. For AI to be aligned, it must have compatible values or goals with some set of humans, who may themselves have conflicting goals. The same AI system can be aligned with a set of stakeholders in one domain and misaligned with them in another. Judging whether AI models are aligned thus depends on who is included in the assessment and what they care about (Kierans et al., 2025). Thus, technical progress in areas like preference aggregation or value learning provides only some of what alignment requires.

Pluralistic alignment is a neverending process. Furthermore, human preferences evolve over time. As a result, any approach to solving alignment that frames the problem as one about specifying the right behaviors and objectives makes a category error and risks entrenching existing power structures. It is necessary to design approaches to aligning AI which recognize that “deliberation should be the goal of AI Safety, not just the procedure by which it is ensured” (Dobbe et al., 2021).

4. Structural solutions

4.1. Ensure that alignment research actually advances alignment (and not just profit)

The institutional barriers above imply that no increase in research throughput can produce alignment in the relevant sense without infrastructure that gives alignment a legitimate target. Meaningful alignment requires addressing both “the technical deficit of conventional regulation and the democratic deficit of industry standards” (Hadfield & Clark, 2023): the status quo of industry making value-laden alignment decisions is not democratically accountable. Closing these gaps requires deliberative processes whose outputs

can shape the evaluation regimes maintained by AI Safety Institutes and the third-party auditing ecosystem.

4.2. Ensure that alignment research is actually accelerated (and not just capabilities)

Protect compute and evaluation authority. Voluntary safety commitments have a poor track record. OpenAI committed 20% of compute to its Superalignment team in July 2023, but “four people who worked on or closely with the team said that the actual resources were between one and two per cent of the company’s compute,” and the team was dissolved within a year (Ronan Farrow & Andrew Marantz, 2026). **Structurally independent oversight** would replace these discretionary commitments with binding ones: a fixed fraction of training compute reserved for designated evaluators, combined with prespecified authority to delay deployments that fail safety criteria. Without such authority, automated alignment outputs and the compute notionally allocated to them remain at the discretion of the parties they were meant to constrain.

Protect career incentives. Researchers face a dilemma between access and independence: working inside frontier labs offers compute and influence at the cost of intellectual independence, while external positions sacrifice the access needed to study deployed systems. **Independent research career tracks**, supported by multi-year grants evaluated by conceptual contribution rather than benchmark performance, would provide institutional homes for the work that commercial centers consistently fail to support.

5. Conclusion

The assumption that beneficial AI alignment research is capability-bottlenecked offers a tempting resolution to the discomfort of building in the present under uncertainty about the future. If AI will solve alignment in the critical window, the structural problems identified here become less urgent—to be handled when the time comes, by systems more capable than we are. This assumption also offers a very convenient agenda for companies who are racing to develop increasingly powerful AI.

We have argued that this is not realistic. The parts of alignment that remain unsolved are not waiting for smarter, more numerous researchers; they depend on whether we build mechanisms that allow accelerated research to accumulate into something reliable. That is not yet happening, at least to public knowledge. Absent meaningful effort to build this infrastructure, “automated alignment” might be abused as a safety-washed euphemism for automating AI capabilities progress. However, we hope that the solutions described above could provide a pathway to accelerate beneficial alignment research, so long as AI developers recognize the ur-

gency of structural investments despite their slow returns.

Impact Statement

This paper argues that catastrophic risks from advanced AI cannot be adequately mitigated through automated alignment research alone. By challenging the prevalent claim that automating alignment research is sufficient, we hope to redirect attention and resources toward the institutional infrastructure upon which beneficial AI ultimately depends. A potential risk of our argument is that it could be read as discounting the value of ongoing technical alignment work; that is not our intention, and we emphasize that technical and institutional progress are complements rather than substitutes.

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